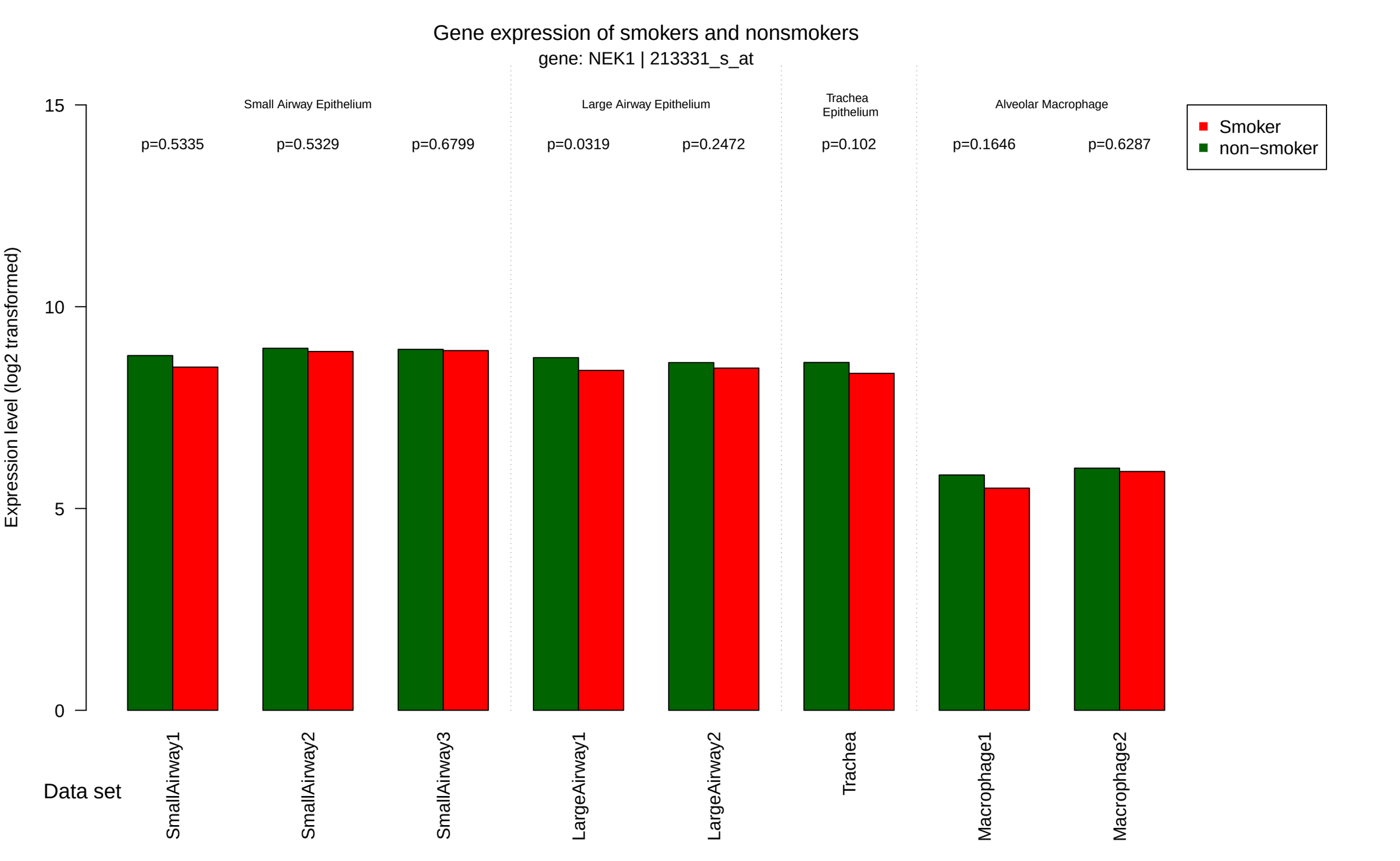
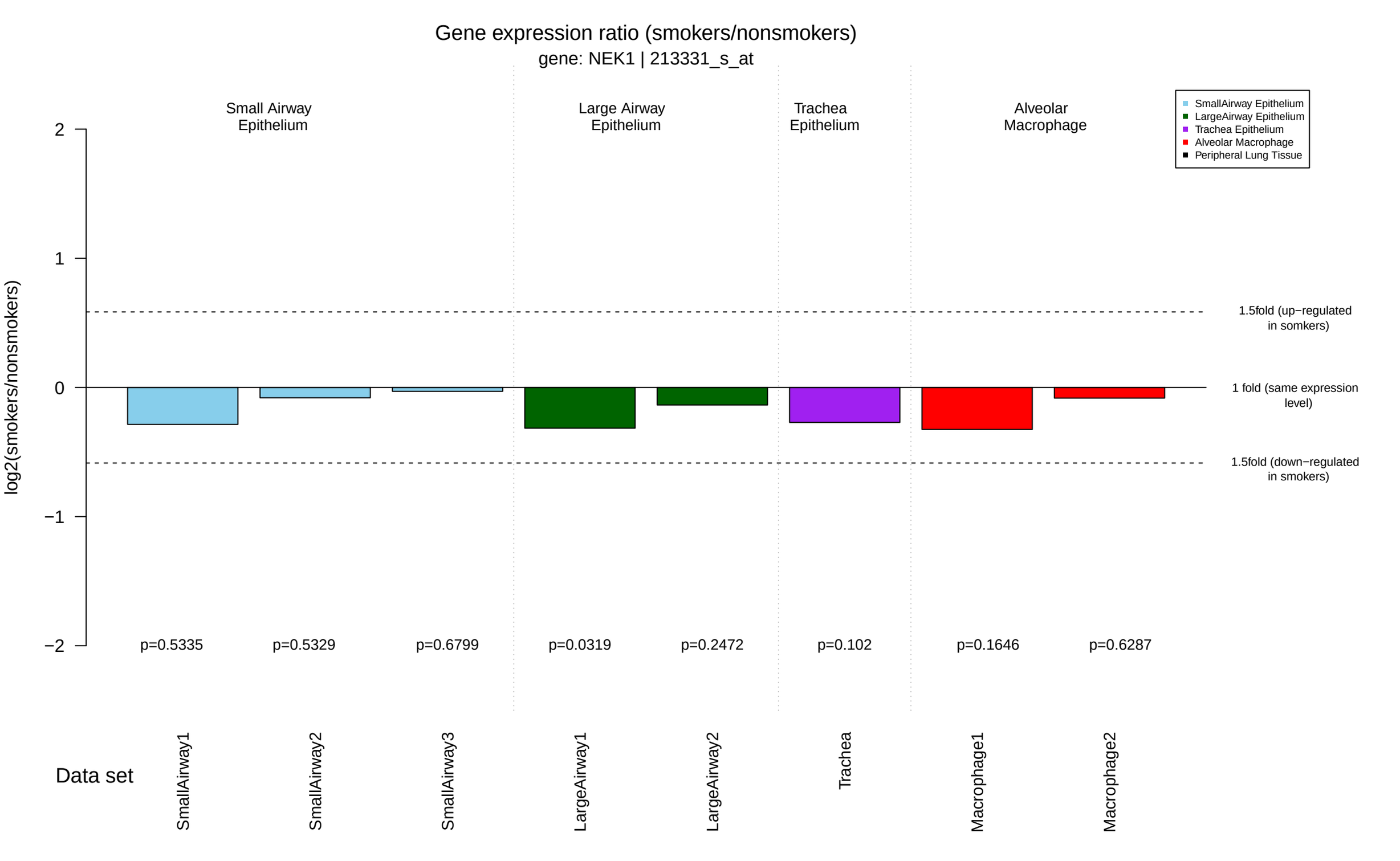
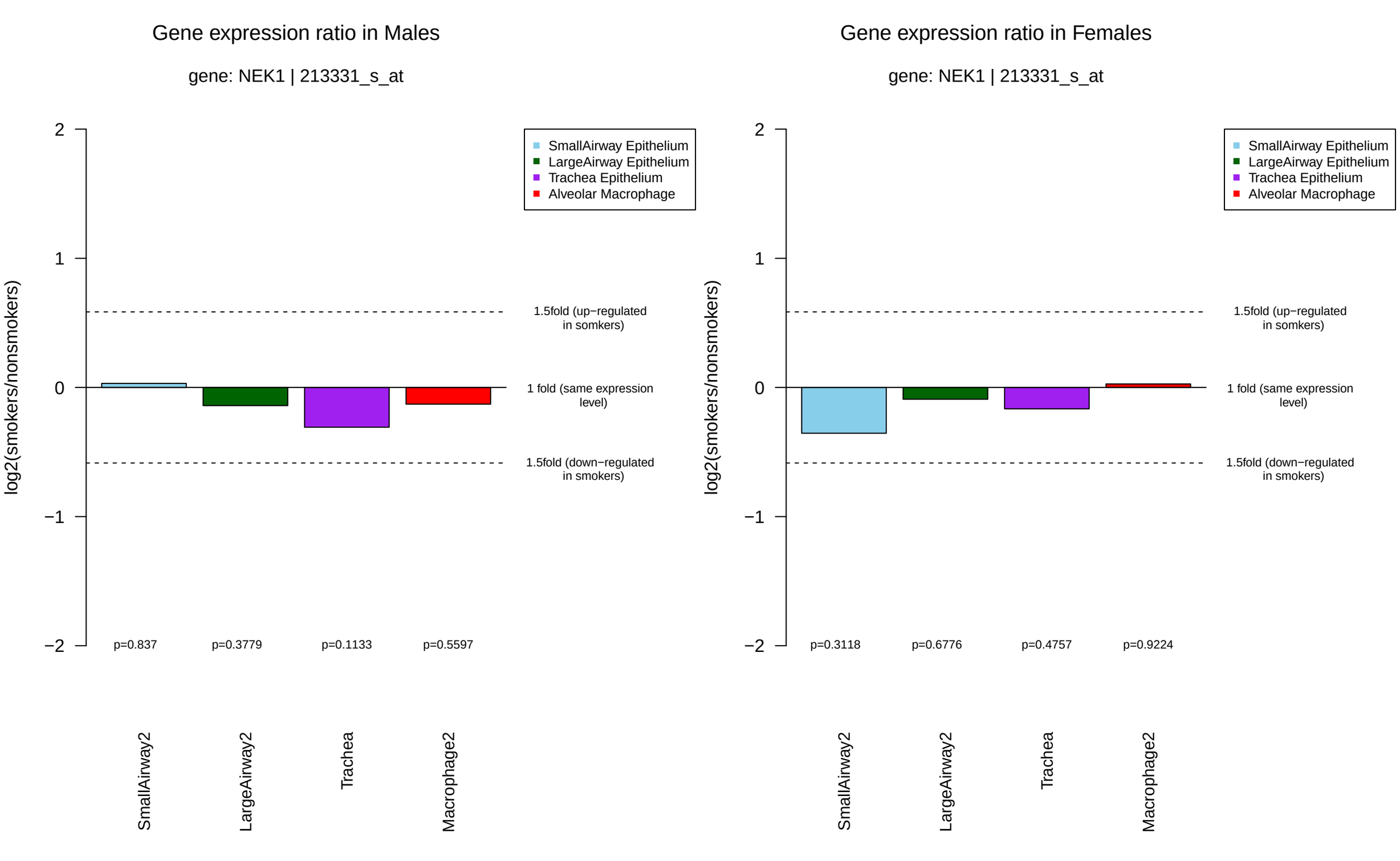
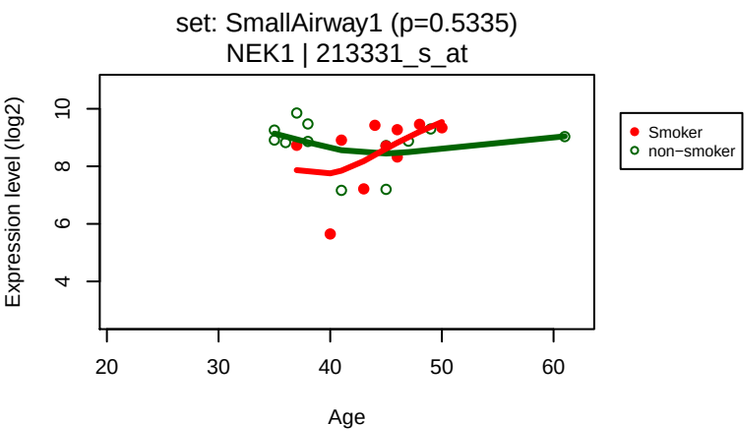
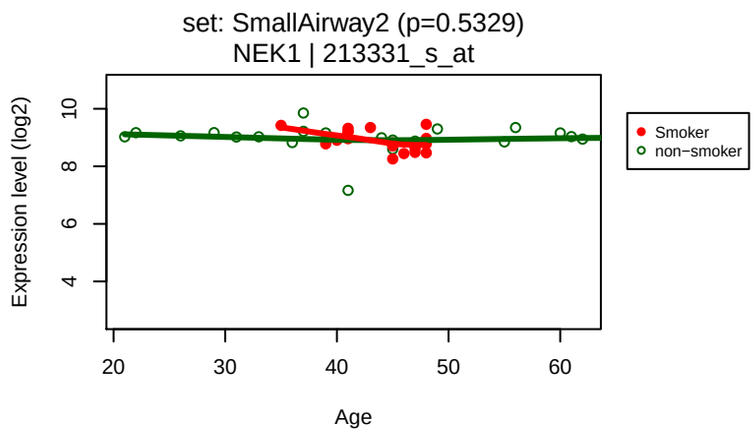
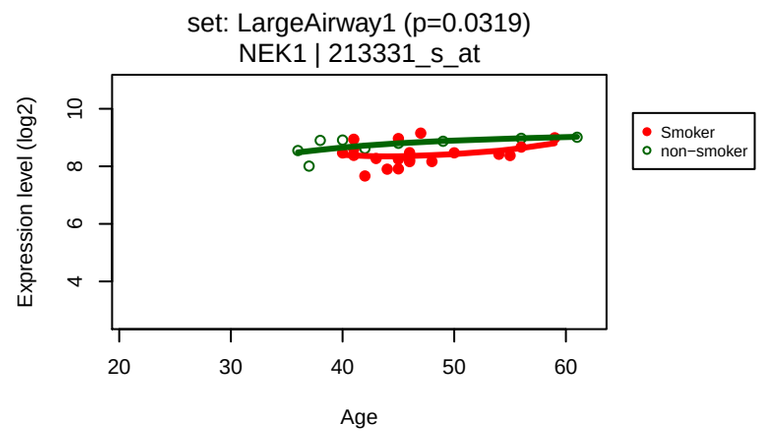

Gene expression of smokers and nonsmokers
gene: NEK1 | 213331_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5335
p=0.5329
p=0.6799
p=0.0319
p=0.2472
p=0.102
p=0.1646
p=0.6287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

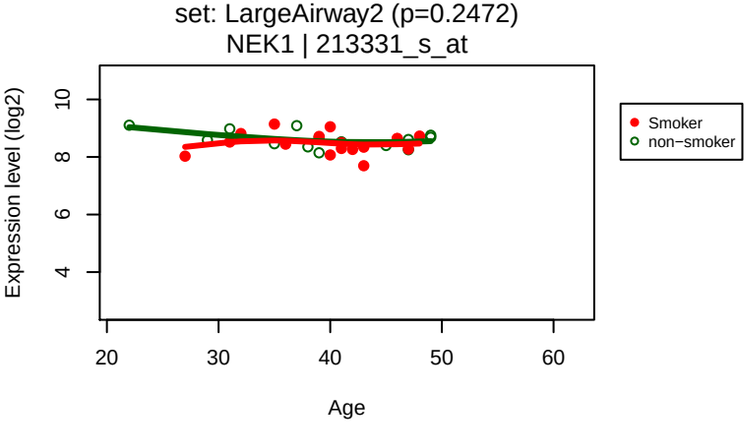

Gene expression ratio (smokers/nonsmokers)
gene: NEK1 | 213331_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5335
p=0.5329
p=0.6799
p=0.0319
p=0.2472
p=0.102
p=0.1646
p=0.6287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

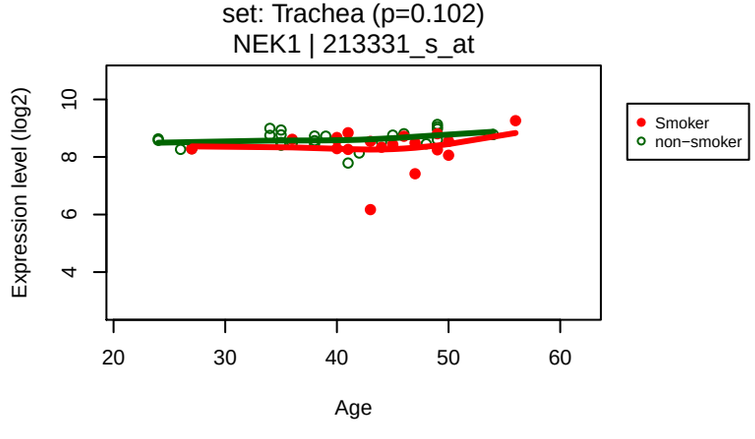

Gene expression ratio in Males
gene: NEK1 | 213331_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.837
p=0.3779
p=0.1133
p=0.5597
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK1 | 213331_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3118
p=0.6776
p=0.4757
p=0.9224
SmallAirway2
LargeAirway2
Trachea
Macrophage2

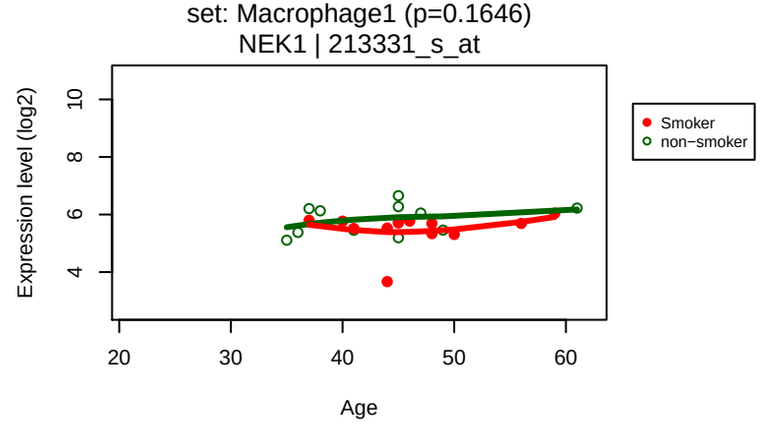

set: SmallAirway1 (p=0.5335)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker

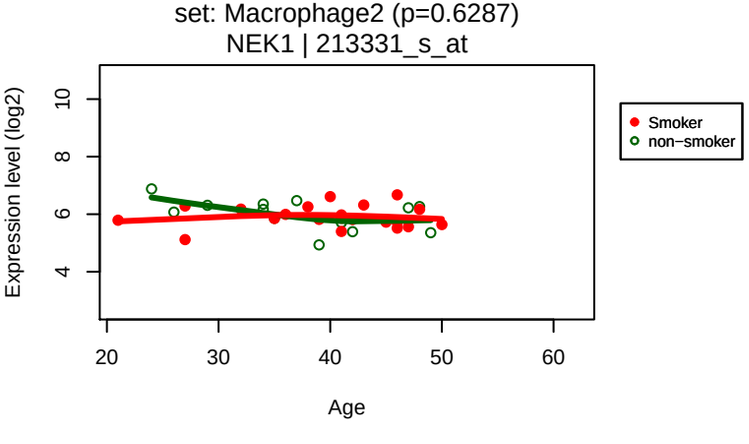

set: SmallAirway2 (p=0.5329)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker

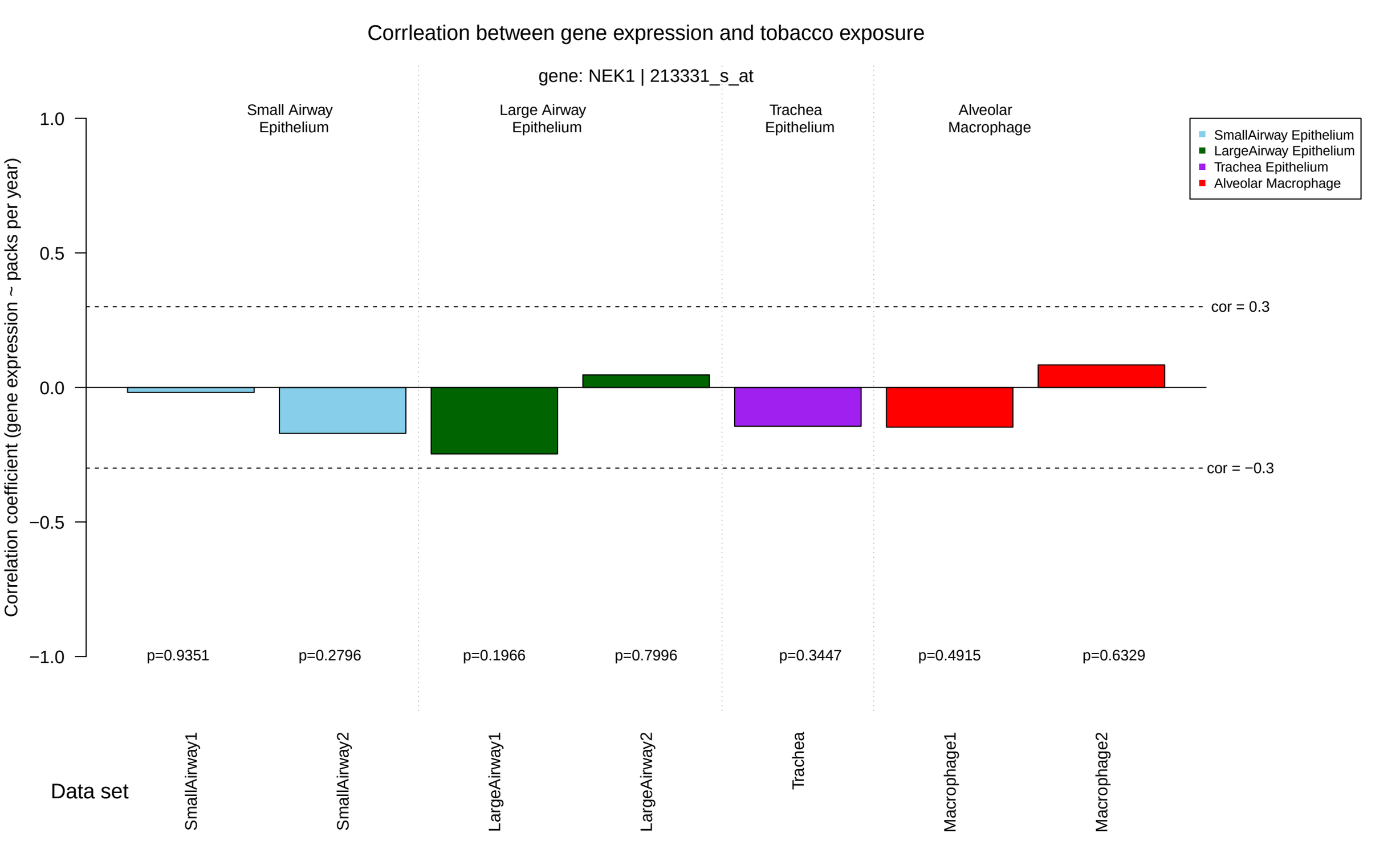

set: LargeAirway1 (p=0.0319)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2472)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.102)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1646)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6287)
NEK1 | 213331_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK1 | 213331_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9351
p=0.2796
p=0.1966
p=0.7996
p=0.3447
p=0.4915
p=0.6329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2